

Coursework

Due Date: [03/01/2025](#)

1

Problem 1

Answer: (a)**Proof:** To prove that the function

$$f(x) = \max\{ax + b, x^4 - 5, e^{-x}\}$$

is convex, we will use the fact that the pointwise maximum of convex functions is convex, provided each function involved is convex.

1. Linear Function $ax + b$:

A linear function is both convex and concave. Its second derivative is zero, which satisfies the condition for convexity.

$$\frac{d^2}{dx^2}(ax + b) = 0 \geq 0$$

2. Polynomial Function $x^4 - 5$:

To check convexity, compute the second derivative:

$$\frac{d}{dx}(x^4 - 5) = 4x^3$$

$$\frac{d^2}{dx^2}(x^4 - 5) = 12x^2$$

Since $12x^2 \geq 0$ for all $x \in \mathbb{R}$, the function $x^4 - 5$ is convex.

3. Exponential Function e^{-x} :

Compute the second derivative:

$$\frac{d}{dx}e^{-x} = -e^{-x}$$

$$\frac{d^2}{dx^2}e^{-x} = e^{-x}$$

Since $e^{-x} > 0$ for all $x \in \mathbb{R}$, the function e^{-x} is convex.

Given that each of the functions $ax + b$, $x^4 - 5$, and e^{-x} is convex, the pointwise maximum of these functions is also convex.

Therefore,

$$f(x) = \max\{ax + b, x^4 - 5, e^{-x}\}$$

is a convex function.

□

Problem 2

Answer: (a)

The subdifferential of a function $f(x)$ at a point x_0 , denoted $\partial f(x_0)$, is the set of all subgradients $g \in \mathbb{R}$ such that

$$f(x) \geq f(x_0) + g(x - x_0), \quad \forall x \in \text{dom}(f).$$

Case 1: $x \in (-1, 0)$

For $x \in (-1, 0)$, the function is $f(x) = -x$. Since $f(x)$ is differentiable in this interval, the subdifferential contains only the derivative:

$$\frac{df}{dx} = -1.$$

Thus,

$$\partial f(x) = \{-1\}, \quad x \in (-1, 0).$$

Case 2: $x = 0$

At $x = 0$, the function $f(x)$ is continuous but not differentiable. We calculate the subdifferential by considering both left and right limits.

- For $x \in (-1, 0]$, $f(x) = -x$, so the left-hand derivative at $x = 0$ is:

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x - 0}{x} = -1.$$

- For $x \geq 0$, $f(x) = x^2$, so the right-hand derivative at $x = 0$ is:

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x^2 - 0}{x} = 0.$$

Since $f(x)$ is convex, the subdifferential at $x = 0$ is the interval between these two slopes:

$$\partial f(0) = [-1, 0].$$

Case 3: $x > 0$

For $x > 0$, the function is $f(x) = x^2$. Since $f(x)$ is differentiable in this interval, the subdifferential contains only the derivative:

$$\frac{df}{dx} = 2x.$$

Thus,

$$\partial f(x) = \{2x\}, \quad x > 0.$$

So (a) is the answer.

Problem 3

Answer: (a)

1. First Term: $\langle Ax, x \rangle$:

Recall that $\langle Ax, x \rangle = x^\top Ax$, a quadratic form. The gradient of this term with respect to x is:

$$\nabla_x \langle Ax, x \rangle = (A + A^*)x.$$

2. Second Term: $\langle x, b \rangle$:

The term $\langle x, b \rangle = x^\top b$ is linear. The gradient with respect to x is:

$$\nabla_x \langle x, b \rangle = b.$$

The gradient of $f(x)$ is the sum of the gradients of its terms:

$$\nabla_x f(x) = (A + A^*)x + b.$$

Problem 4

Answer: (a)

Given the definition of Fenchel conjugate:

$$f^*(u) = \sup_{x \in \mathbb{R}} (u^\top x - f(x)).$$

Since we have $f(x) = g(3x)$,

$$f^*(u) = \sup_{x \in \mathbb{R}} (u^\top x - g(3x)) = \sup_{3x \in \mathbb{R}} \left(\frac{1}{3} u^\top 3x - g(3x) \right) = \sup_{y \in \mathbb{R}} \left(\frac{1}{3} u^\top y - g(y) \right) = g^* \left(\frac{u}{3} \right).$$

2

Problem 1

(1)

$$f^*(u) = \sup_{x > 0} (\langle x, u \rangle + \log(x))$$

When $x > 0$ we have:

$$f'(x) = -\frac{1}{x} =: u \quad \Rightarrow \quad x = -\frac{1}{u} \quad u < 0$$

$$f(x)|_{-\frac{1}{u}} = \log(-u)$$

So the tangent line of $-\log x$ is

$$\ell(x) = u \left(x + \frac{1}{u} \right) + \log(-u)$$

We have

$$f^*(u) = -\ell(0) = -1 - \log(-u)$$

When $u > 0$, we have $f^*(u) = +\infty$. So in total:

$$f^*(u) = \begin{cases} +\infty & u \geq 0 \\ -1 - \log(-u) & u < 0 \end{cases}$$

(2)

$$f^*(u) = \sup_{x>0} (\langle x, u \rangle - 2x^2)$$

$$f'(x) = 4x =: u \quad x = \frac{1}{4}u$$

$$f(x)|_{\frac{1}{4}u} = \frac{1}{8}u^2$$

The tangent line is

$$\ell(x) = u(x - \frac{1}{4}u) + \frac{1}{8}u^2$$

We have

$$f^*(u) = -\ell(0) = \frac{1}{8}u^2$$

(3)

$$\iota_{[-1,1]}^*(u) = \sup_{x \in [-1,1]} \langle x, u \rangle = \sup_{x \in [-1,1]} |x||u| = |u|$$

The second equality holds because of Cauchy Schwartz inequality and equality can be attained.

Problem 2

(1)

Proof: It is trivial for $n = 1$ case such that

$$f(\lambda_1 x_1) = f(x_1) = \lambda_1 f(x_1)$$

Suppose Jensen's inequality holds for $n = N$:

$$f\left(\sum_{i=1}^N \lambda_i x_i\right) \leq \sum_{i=1}^N \lambda_i f(x_i) \quad \sum_{i=1}^N \lambda_i = 1$$

We then have:

$$\sum_{i=1}^{N+1} \lambda'_i x_i = (1 - \lambda'_{N+1}) \sum_{i=1}^N \lambda_i x_i + \lambda'_{N+1} x_{N+1}$$

where

$$\lambda'_i = (1 - \lambda'_{N+1})\lambda_i \quad \text{for } i = 1, \dots, N$$

and

$$\sum_{i=1}^{N+1} \lambda'_i = (1 - \lambda'_{N+1}) \sum_{i=1}^N \lambda_i + \lambda'_{N+1} = 1$$

So we have:

$$\begin{aligned} f\left(\sum_{i=1}^{N+1} \lambda'_i x_i\right) &= f\left((1 - \lambda'_{N+1}) \sum_{i=1}^N \lambda_i x_i + \lambda'_{N+1} x_{N+1}\right) \leq (1 - \lambda'_{N+1}) f\left(\sum_{i=1}^N \lambda_i x_i\right) + \lambda'_{N+1} f(x_{N+1}) \\ &\leq (1 - \lambda'_{N+1}) \sum_{i=1}^N \lambda_i f(x_i) + \lambda'_{N+1} f(x_{N+1}) \\ &= \sum_{i=1}^{N+1} \lambda'_i f(x_i) \end{aligned}$$

□

(2)

$-\log$ is twice differentiable. We can check its second order derivative.

Proof: Let $g(x) := -\log(x)$,

$$\text{dom } g = (0, +\infty)$$

$$g''(x) = \frac{1}{x^2} > 0$$

So we have:

$$g''(x)v^2 > 0 \quad \forall v \in (0, +\infty), \forall x \in \text{dom } g$$

Hence $g(x)$ is convex.

□

(3)

Proof:

$$\begin{aligned} -\log \sqrt[n]{x_1 \cdots x_n} &= -\log \left(x_1^{\frac{1}{n}} \cdots x_n^{\frac{1}{n}}\right) \\ &= -\frac{1}{n} \sum_{i=1}^n \log x_i \\ &\geq -\log \left(\frac{1}{n} \sum_{i=1}^n x_i\right) \quad \text{by Jensen's inequality} \end{aligned}$$

Take opposite sign and the exponential on both side, we have proved the arithmetic-geometric inequality. □

Problem 3

Proof: Let $A := \{a_1, \dots, a_m\}$, we have polytope:

$$C = \text{co}(A) := \left\{ \sum_{i=1}^m \lambda_i a_i \mid \lambda_i \geq 0 \text{ for all } i, \sum_{i=1}^m \lambda_i = 1 \right\}$$

For $\forall x \in C, x = \sum_{i=1}^m \lambda_i a_i$ and a convex function f , we have:

$$\begin{aligned} f(x) &= f\left(\sum_{i=1}^m \lambda_i a_i\right) \leq \sum_{i=1}^m \lambda_i f(a_i) \quad \text{by Jensen's inequality} \\ &\leq \sum_{i=1}^m \lambda_i \max_i f(a_i) \\ &\leq \max_i f(a_i) \end{aligned}$$

□

Problem 4

Proof: For $\forall \lambda \in [0, 1]$, we have:

$$\begin{aligned} f(\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)) &= \|2(\lambda x_1 + (1 - \lambda)x_2) - (\lambda y_1 + (1 - \lambda)y_2)\|_2^2 \\ &= \|\lambda(2x_1 - y_1) + (1 - \lambda)(2x_2 - y_2)\|_2^2 \\ &= \|\lambda(2x_1 - y_1)\|_2^2 + \|(1 - \lambda)(2x_2 - y_2)\|_2^2 \\ &\quad + 2\lambda(1 - \lambda)\langle 2x_1 - y_1, 2x_2 - y_2 \rangle \\ &\leq \lambda^2\|2x_1 - y_1\|_2^2 + (1 - \lambda)^2\|2x_2 - y_2\|_2^2 \\ &\quad + 2\lambda(1 - \lambda)\|2x_1 - y_1\|_2\|2x_2 - y_2\|_2 \\ &\leq \lambda^2\|2x_1 - y_1\|_2^2 + (1 - \lambda)^2\|2x_2 - y_2\|_2^2 \\ &\quad + \lambda(1 - \lambda)(\|2x_1 - y_1\|_2^2 + \|2x_2 - y_2\|_2^2) \\ &= \lambda\|2x_1 - y_1\|_2^2 + (1 - \lambda)\|2x_2 - y_2\|_2^2 \\ &= \lambda f(x_1, y_1) + (1 - \lambda)f(x_2, y_2) \end{aligned}$$

The first inequality is due to Cauchy Schwartz. The second inequality holds due to $2\|x\|\|y\| \leq \|x\|^2 + \|y\|^2$. □

Problem 5

(1)

We denote $\overline{B}_\infty(b, \epsilon)$ as the closed inf norm ball centred at b with radius ϵ . The primal problem can be written as:

$$\min_x \underbrace{\frac{1}{2}\|x\|^2}_{f(x)} + \underbrace{\iota_{\overline{B}_\infty(b, \epsilon)}(Ax)}_{g(Ax)}$$

Use Fenchel-Rockafellar duality theory, we have:

$$\begin{aligned} & \min_u \iota_{\overline{B}_\infty(b, \varepsilon)}^*(u) + \frac{1}{2} \|A^\top u\|^2 \\ \implies & \min_u \sigma_{\overline{B}_\infty(b, \varepsilon)}(u) + \frac{1}{2} \|A^\top u\|^2 \end{aligned}$$

where $\sigma_C(u)$ is the support function of C . We also have:

$$Ax \in \overline{B}_\infty(b, \varepsilon) \iff Ax \in b + \varepsilon B_\infty$$

$$\begin{aligned} \sigma_{\overline{B}_\infty(b, \varepsilon)}(u) &= \varepsilon \sup_{Ax' \in B_\infty} \langle Ax', u \rangle + \langle b, u \rangle \\ &= \varepsilon \|u\|_1 + \langle b, u \rangle \end{aligned}$$

So the dual problem is:

$$\min_u \varepsilon \|u\|_1 + \langle b, u \rangle + \frac{1}{2} \|A^\top u\|^2$$

(2)

$$\begin{aligned} \text{dom } f &= \mathbb{R}^d, \\ \text{dom } g &= \overline{B}_\infty(b, \varepsilon). \end{aligned}$$

For QC (qualification condition), we need to have:

$$0 \in \text{int}(\text{dom } g - A \text{dom } f)$$

$$\iff 0 \in \text{int}\{u - Ax \mid u \in \text{dom } g, x \in \text{dom } f\}$$

If $\text{rank}(A) = n$, $\text{col}(A)$ spans $\mathbb{R}^n$:

$$\exists x_0 \text{ s.t. } Ax_0 = b$$

g is continuous at Ax_0 , QC holds.

If $\text{rank}(A) < n$, we set:

$$\delta := \inf_{x \in \mathbb{R}^d} \|Ax - b\|_\infty.$$

Then:

If $\delta < \varepsilon$, g is continuous at $Ax \quad \forall x \text{ s.t. } \|Ax - b\|_\infty < \varepsilon$, QC holds.

If $\delta = \varepsilon$, for $Ax \in \text{dom } g$, they all locate on the boundary of $\text{dom } g$. Since g is continuous on the boundary, QC holds.

If $\delta > \varepsilon$, No feasible set, the problem is not solvable.

To sum, as long as the question has feasible constraints, QC holds and it has strong duality.

(3)

The KKT conditions are:

$$\begin{aligned} -A^\top u &\in \partial f(x) \quad \text{and} \quad u \in \partial g(Ax) \\ \iff x &\in \partial f^*(-A^\top u) \quad \text{and} \quad Ax \in \partial g^*(u) \end{aligned}$$

Since we have:

$$\begin{aligned} \partial f(x) &= \{x\} \\ \partial g(Ax) &= N_{\overline{B}_\infty(b, \varepsilon)}(Ax) \\ \partial f^*(-A^\top u) &= \{-A^\top u\} \\ \partial g^*(u) &= \begin{cases} [b - \varepsilon, b + \varepsilon] & u_i = 0 \quad \forall i = 1, 2, \dots, n \\ \{\varepsilon \text{sign}(u_i) + b\} & \text{otherwise} \end{cases} \end{aligned}$$

where $N_{\overline{B}_\infty(b, \varepsilon)}(Ax)$ is the normal cone at $Ax \in \overline{B}_\infty(b, \varepsilon)$ such that

$$N_{\overline{B}_\infty(b, \varepsilon)}(Ax) = \left\{ u \in \mathbb{R}^n \mid \langle u, y - Ax \rangle \leq 0 \quad \forall y \in \overline{B}_\infty(b, \varepsilon) \right\}$$

For $\|Ax - b\|_\infty < \varepsilon$,

$$N_{\overline{B}_\infty(b, \varepsilon)}(Ax) = \{\mathbf{0}\}$$

For $\|Ax - b\|_\infty = \varepsilon$, we must have $|(Ax)_i - b_i| = \varepsilon$ for some i . We denote:

$$I(Ax) = \{i \mid |(Ax)_i - b_i| = \varepsilon\}$$

We also have:

$$y \in \overline{B}_\infty(b, \varepsilon) \implies \|y - b\|_\infty \leq \varepsilon$$

For $i \in I(Ax)$, if $(Ax)_i = b_i + \varepsilon$,

$$-2\varepsilon \leq y_i - (Ax)_i \leq 0$$

Since $\langle u, y - Ax \rangle \leq 0 \quad \forall y \in \overline{B}_\infty(b, \varepsilon)$, we have to enforce $u_i \geq 0$.

If $(Ax)_i = b_i - \varepsilon$,

$$0 \leq y_i - (Ax)_i \leq 2\varepsilon$$

which implies $u_i \leq 0$.

For $i \notin I(Ax)$, $y_i - (Ax)_i$ can be positive or negative, so we can only enforce $u_i = 0$.

In sum, we have:

$$N_{\overline{B}_\infty(b,\varepsilon)}(Ax) = \left\{ u \in \mathbb{R}^n \mid \begin{cases} u_i((Ax)_i - b_i) \geq 0 & \text{if } |(Ax)_i - b_i| = \varepsilon, \\ u_i = 0 & \text{if } |(Ax)_i - b_i| < \varepsilon, \end{cases} \quad \forall i = 1, 2, \dots, n \right\}$$

From $Ax \in \partial g^*(u)$, we have:

$$\begin{cases} |(Ax)_i - b_i| \leq \varepsilon & \text{if } u_i = 0, \\ (Ax)_i - b_i = \varepsilon & \text{if } u_i > 0, \\ (Ax)_i - b_i = -\varepsilon & \text{if } u_i < 0, \end{cases} \quad \forall i = 1, 2, \dots, n$$

The KKT conditions become:

$$\begin{aligned} -A^\top u = x \quad \text{and} \quad & \begin{cases} u_i((Ax)_i - b_i) \geq 0 & \text{if } |(Ax)_i - b_i| = \varepsilon, \\ u_i = 0 & \text{if } |(Ax)_i - b_i| < \varepsilon, \end{cases} \quad \forall i = 1, 2, \dots, n \\ \iff -A^\top u = x \quad \text{and} \quad & \begin{cases} |(Ax)_i - b_i| \leq \varepsilon & \text{if } u_i = 0, \\ (Ax)_i - b_i = \varepsilon & \text{if } u_i > 0, \\ (Ax)_i - b_i = -\varepsilon & \text{if } u_i < 0, \end{cases} \quad \forall i = 1, 2, \dots, n \end{aligned}$$

(4)

The dual problem is:

$$\begin{aligned} \min_u \varepsilon \|u\|_1 + \langle u, b \rangle + \frac{1}{2} \|A^\top u\|^2 &=: \psi(u) \\ \implies \min_u \varepsilon \|u\|_1 + \langle u, b \rangle + \frac{1}{2} u^\top A A^\top u. \end{aligned}$$

FISTA:

$$\begin{aligned} u_{k+1} &= \text{prox}_{\gamma g^*}(v_k + \gamma A \nabla f^*(-A^\top v_k)), \\ v_{k+1} &= u_{k+1} + \frac{t_k - 1}{t_{k+1}}(u_{k+1} - u_k). \end{aligned}$$

$\implies$

$$\begin{aligned} u_{k+1} &= \text{prox}_{\gamma g^*}(v_k - \gamma A A^\top v_k) = \text{prox}_{\gamma g^*}((I_n - \gamma A A^\top) v_k), \\ v_{k+1} &= u_{k+1} + \frac{t_k - 1}{t_{k+1}}(u_{k+1} - u_k). \end{aligned}$$

where

$$\text{prox}_{\gamma g^*} = (I_n + \partial(\gamma g^*))^{-1}$$

$$\partial(\gamma g^*)(u) = \begin{cases} [\gamma(b - \varepsilon), \gamma(b + \varepsilon)] & \text{if } u_i = 0, \\ \{\gamma(\varepsilon \operatorname{sign}(u_i) + b)\} & \text{otherwise.} \end{cases}$$

$$\operatorname{prox}_{\gamma g^*}(u) = \begin{cases} 0 & \gamma(b - \varepsilon) \leq u_i \leq \gamma(b + \varepsilon), \\ u_i - \gamma(b - \varepsilon) & u_i < \gamma(b - \varepsilon), \\ u_i - \gamma(b + \varepsilon) & u_i > \gamma(b + \varepsilon). \end{cases}$$

Since

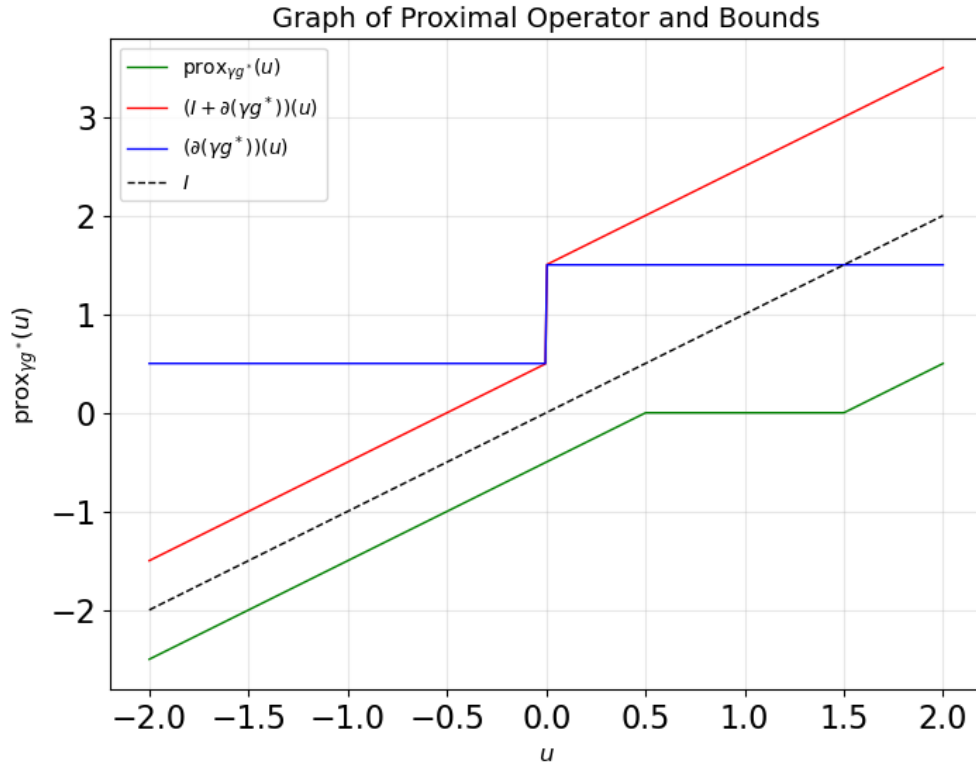


Figure 1.1. Graphical representation of $\operatorname{prox}_{\gamma g^*}$.

$$f(x) = \frac{1}{2}\|x\|^2 \quad \text{is 1-strongly convex}$$

as

$$\nabla^2 f(x) = I, \quad \lambda_{\min}(\nabla^2 f(x)) = 1.$$

This means f^* is 1-Lipschitz smooth.

$$f^* \circ (-A^\top) \quad \text{is Lipschitz smooth with constant } \|A\|^2,$$

so choosing

$$\gamma \leq 1/\|A\|^2.$$

and by FISTA and convergence property of proximal, we have:

$$\psi(u_k) - \min \psi \leq \frac{\|v_0 - \bar{u}\|^2}{2\gamma t_{k+1}^2} = \mathcal{O}\left(\frac{1}{k^2}\right).$$

by choosing

$$t_k = \frac{1 + \sqrt{1 + 4t_{k-1}^2}}{2}$$

so that

$$\frac{1}{t_{k-1}} \leq \frac{2}{k+1}$$

where

$$\bar{u} = \arg \min_u \psi(u).$$

We have:

$$\bar{x} = \nabla f^*(-A^\top \bar{u}) = -A^\top \bar{u},$$

and

$$x_k = \nabla f^*(-A^\top u_k) = -A^\top u_k.$$

Finally we have:

$$\frac{1}{2}\|x_k - \bar{x}\|^2 \leq \psi(u_k) - \min \psi = \mathcal{O}\left(\frac{1}{k^2}\right),$$

$$\|x_k - \bar{x}\| = \mathcal{O}\left(\frac{1}{k}\right).$$

So the objective gap in the dual is at rate $\mathcal{O}\left(\frac{1}{k^2}\right)$ and distance to the primal solution is at rate $\mathcal{O}\left(\frac{1}{k}\right)$.

3

```

1 import numpy as np
2 from scipy import linalg as scplinalg
3 import matplotlib as mpl
4 import matplotlib.pyplot as plt
5 from matplotlib import cm
6
7 import random
8
9 # increase font size for more readable plots
10 fontsize = 18
11 plt.rcParams.update({'axes.labelsize': fontsize,
12                      'font.size': fontsize,
13                      'legend.fontsize': fontsize,

```

```

14         'xtick.labelsize': fontsize - 2,
15         'ytick.labelsize': fontsize - 2})
16 %matplotlib inline
17
18 def generate_problem(n, d, s, std=0.06):
19     # Generate xs
20     # vectors with entries in [0.5, 1] and [-1, -0.5]
21     # repectively
22     assert s % 2 == 0, "s needs to be divisible by 2"
23     xsp = 0.5 * (np.random.rand(s // 2) + 1)
24     xsn = - 0.5 * (np.random.rand(s // 2) + 1)
25     xsparse = np.hstack([xsp, xsn, np.zeros(d - s)])
26     random.shuffle(xsparse)
27     # Generate A
28     A = np.random.randn(n, d)
29     # Generate eps
30     y = A @ xsparse + std * np.random.randn(n)
31     return xsparse, A, y
32
33 n = 1000
34 d = 500
35 s = 50
36 std = 0.06
37
38 np.random.seed(2000)
39 xsparse, A, y = generate_problem(n, d, s, std)

```

Listing 1.1. General setup

(1)

$$\min_{x \in \mathbb{R}^d} \Phi(x) \quad \Phi(x) = \underbrace{\frac{1}{2n} \|Ax - y\|_2^2}_{:=F(x)} + \lambda \|x\|_1 = \frac{1}{n} \sum_{i=1}^n \phi_i(x),$$

where

$$\phi_i(x) = \underbrace{\frac{1}{2} (\langle a^i, x_i \rangle - y_i)^2}_{:=f_i(x)} + \underbrace{\lambda \|x\|_1}_{:=g(x)}$$

```

1 def soft_thresholding(x, rho):
2     """Soft thresholding of scalar x at level rho."""
3     return np.sign(x-rho) * np.maximum(0, np.abs(x) - rho)
4
5 def Phi(x, lambda):
6     return np.linalg.norm(A @ x - y)**2 * 0.5 / n + lambda * np.
7         linalg.norm(x, 1)

```

```

8 grad_F = lambda x: A.T @ (A @ x - y)
9
10 def grad_fi(x, i):
11     return A[i] * (A[i] @ x - y[i])
12
13 def psga(x0, gamma, lmbda, max_iter):
14
15     update = lambda x, i, k: soft_thresholding(x - gamma(k) *
16         grad_fi(x, i), gamma(k) * lmbda)
17
18     x0 = x0
19     xs = [x0]
20     Phis = [Phi(x0, lmbda)]
21     for k in range(max_iter):
22
23         i = np.random.randint(n)
24         x_up = update(xs[-1], i, k)
25
26         xs.append(x_up)
27         Phis.append(Phi(x_up, lmbda))
28
29     xs = np.vstack([x.flatten() for x in xs])
30     Phis = np.vstack(Phis)
31     return xs, Phis

```

Listing 1.2. PSGA

(2)

```

1 def rcpga(x0, gamma, ik, lmbda, max_iter):
2     """ik is a function mapping from an integer
3     k to an index. gamma is an array of size d (or a function of the
4     index i)."""
5
6     def update(x, i):
7         grad = grad_F(x)[i]
8         x_up = x.copy()
9         x_up[i] = soft_thresholding(x[i] - gamma(i) / n * grad,
10             gamma(i) / n * lmbda)
11         return x_up
12
13     ### Run algorithm
14     # Keep values in x, start with initial value x0
15     x0 = x0
16     xs = [x0]
17     Phis = [Phi(x0, lmbda)]

```

```

16
17
18     for k in range(max_iter):
19         i = ik(k)
20         x_up = update(xs[-1], i)
21
22         xs.append(x_up)
23         Phis.append(Phi(x_up, lambda))
24
25     return np.vstack(xs), np.vstack(Phis)

```

Listing 1.3. RCPGA

(3)

We carefully choose $\lambda = 0.1$ with 20000 iterations. The step size in PSGA needs to be scaled as the first several steps are too large to make the algorithm diverge to very large values. So instead of

$$\gamma_k = \frac{n}{\|A\|_2^2 \sqrt{k+1}}$$

we scale to smaller step size

$$\gamma_k = \frac{n}{50\|A\|_2^2 \sqrt{k+1}}$$

We plot the objective function values with number of iterations and sequence of ergodic means. RCPGA converges faster than PSGA, as shown from figure.

```

1 norm_A2 = np.linalg.norm(A, 2)**2
2 gamma_psga = lambda k: n / (50 * norm_A2 * np.sqrt(k + 1))
3 vec_norm_A2 = np.array([n / np.linalg.norm(a)**2 for a in A.T])
4 gamma_rcpga = lambda i: vec_norm_A2[i]
5 i_randomised = lambda k: np.random.randint(d)
6 lambda = 0.1
7 max_iter = 20000
8 x0 = np.zeros(d)
9 x_psga, Phi_psga = psga(x0, gamma_psga, lambda, max_iter)
10 x_rcpga, Phi_rcpga = rcpga(x0, gamma_rcpga, i_randomised, lambda,
    max_iter)
11 gamma_psga_list = np.array([gamma_psga(i) for i in range(max_iter +
    1)])
12 cumsum_gamma = np.cumsum(gamma_psga_list)
13 ergodic_mean = np.cumsum(gamma_psga_list[:, None] * x_psga, axis=0)
    / cumsum_gamma[:, None]
14 Phi_erg_mean = np.array([Phi(x, lambda) for x in ergodic_mean])
15 plt.figure()
16 plt.title('Function Values')
17 plt.plot(Phi_rcpga, label='RCPG')
18 plt.plot(Phi_psga, label='PSG')

```

```
19 plt.plot(Phi_erg_mean, label='Ergodic in PSG')
20 plt.xlabel('# of iteration')
21 plt.legend()
22 plt.show()
```

Listing 1.4. Run Algorithms and Plot

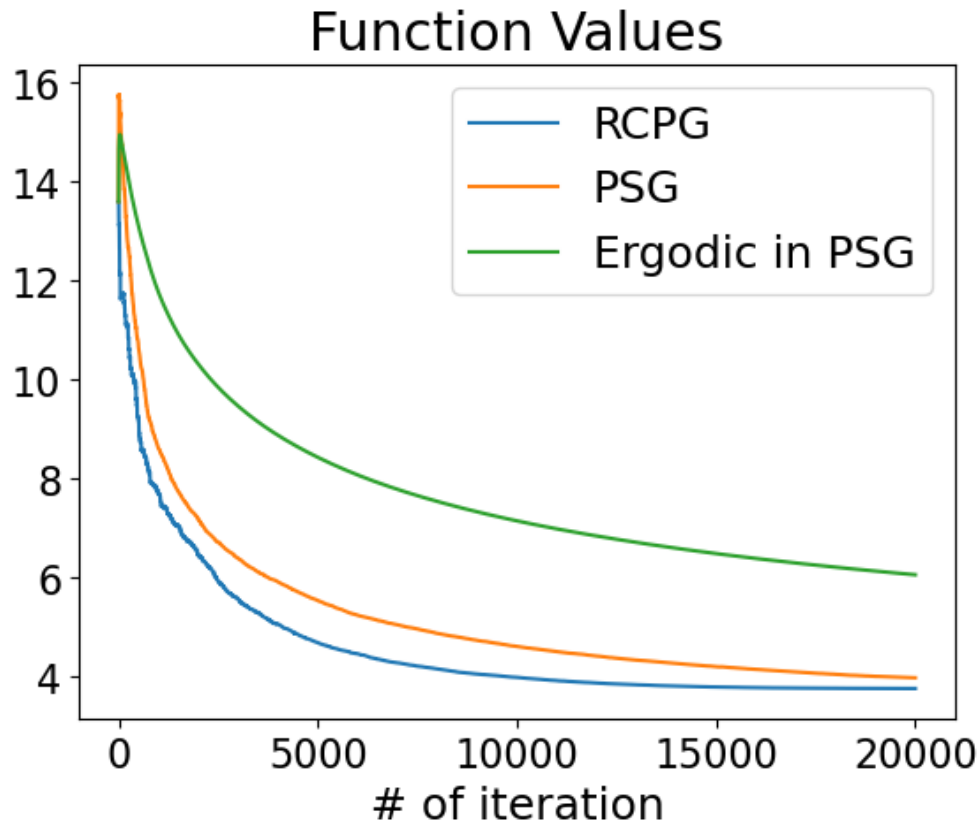


Figure 1.2. PSG v.s. RCPG and Ergodic in PSG